

Bilingual Emotion Processing: The Valence x Arousal Interaction

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**VALENCE
ASYMMETRIES**



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I want to know:

- **Do bilinguals process emotions similarly across their languages?** Are valence and arousal processed the same across both L1 and L2?
- **Which ‘*type*’ of bilinguals are impacted?** How do bilingual demographics impact processing?

The Foreign Language Effect

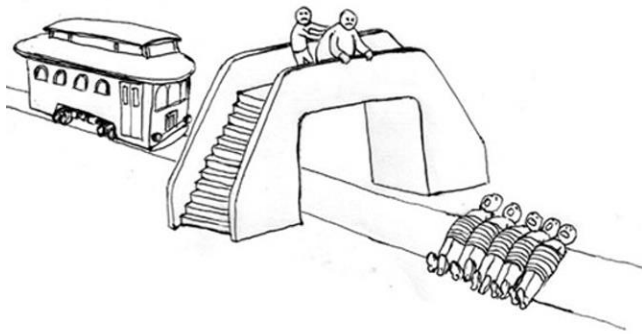
Bilinguals make different decisions when responding in their foreign language:

- Decreased Loss Aversion (Keysar et al., 2012)
- Reduced Causality bias (Diaz-Lago & Matute, 2018)
- Less superstitious (Hadjichristidis et al., 2017)
- Reduces 'winning streak' beliefs (Gao et al., 2015)
- More eco-friendly consumption (less disgust; Geipel et al., 2018)
- Reduced mental imagery (Hayakawa & Keysar, 2018)
- *More sacrificial responding in moral dilemmas...*

The *Moral* Foreign Language Effect

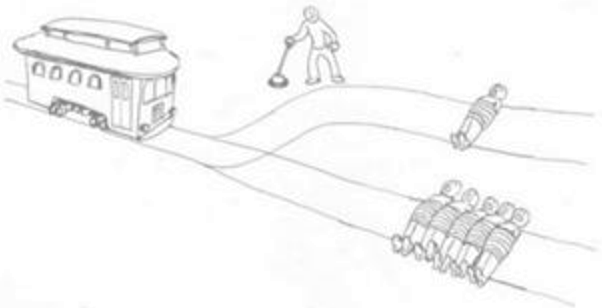
Personal dilemmas = direct actions (e.g., use your hands to push a person)

More emotionally aversive



The footbridge dilemma. A train is going down a track very fast toward five people. The train has a problem and cannot be stopped unless a heavy weight is dropped on the track. There is a very heavy man next to you—your only way to stop the train is to push him onto the track, killing him to save five people. **Would you push him?**

Impersonal dilemmas = indirect actions (e.g., press a button)

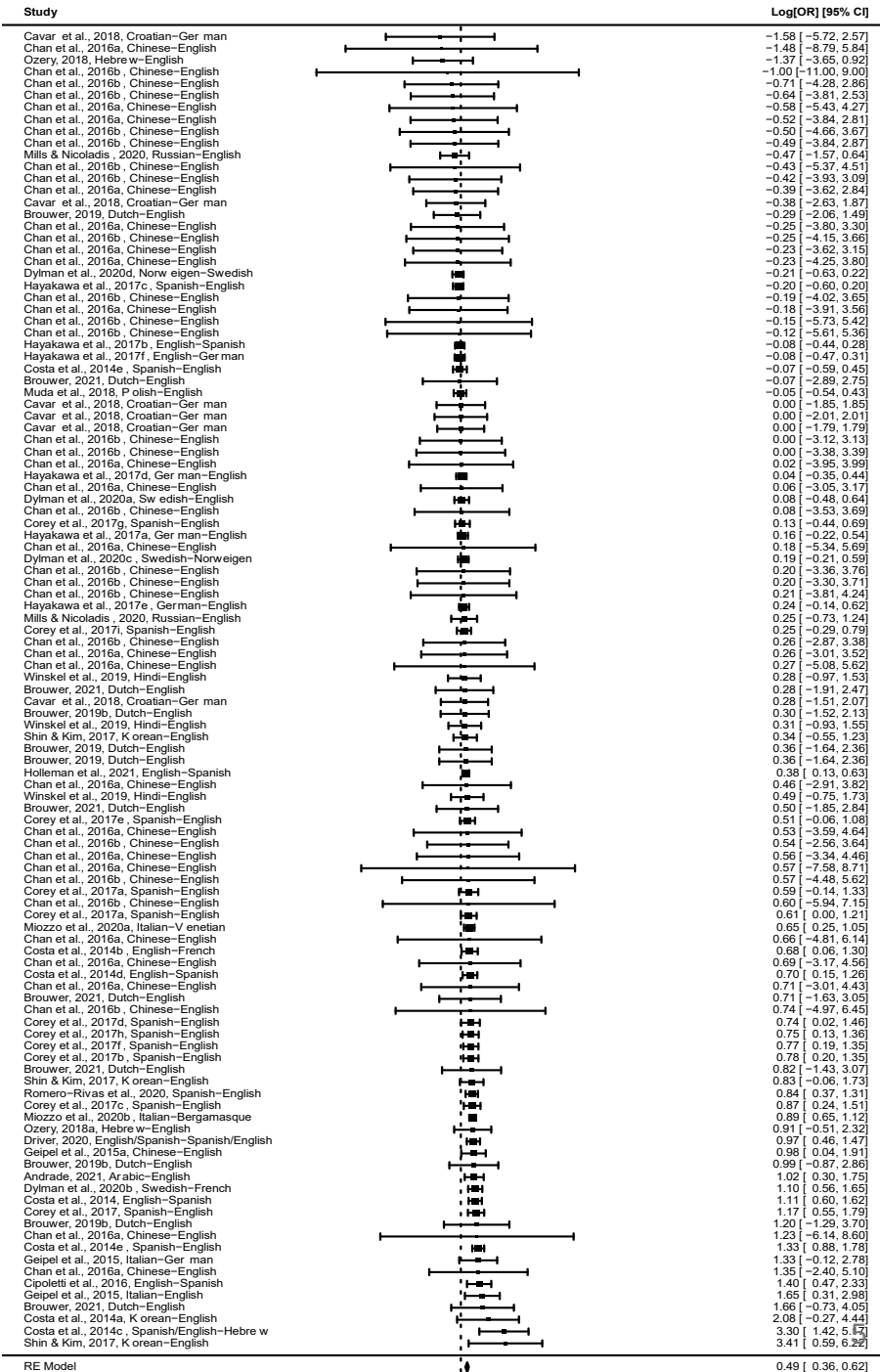


The switch dilemma. A train is going down a track very fast toward five people. The train has a problem and cannot be stopped. Five people will die if you stay on this track. There is another track that you can use to divert the train. At the end of this track there is one man that will die if you change the track. **Would you change the track?**

Meta-analysis

More likely to push:

- Footbridge: **1.98x**
- All personal: **1.64x**
- Moderation: **Lower reading proficiency**
- *Impersonal: no diff, not moderated*
- **Not lower comprehension**
- **“Emotional” task only**



Language & Decision-Making

- *Nativeness* should *not* have an impact outside of comprehension

UNLESS:

- It is a proxy for something else: E.G.: **EMOTION**

BUT

- Competing hypotheses?
 - Cultural differences
 - Increased deliberation
 - Cognitive load

Bilingual Emotion Processing

- Evidence that bilinguals:
 - Experience emotions differently in L2
 - Make different decisions in L2
- The psycholinguistics and cognition literatures are separate fields
 - **Psychology:** Experimental, emotion/cognition
 - **Linguistics:** Lexical factors
 - **Bilingual Research:** Acquisition, individual differences
- Considering all angles is difficult

Problem 1: Bilingual demographics

What type of bilinguals are affected?

- Restrictive sampling (foreign language students)
- Lack of comprehensive measurement - reliance on AOA and self-reported proficiency
- Categorical definitions

Plan to solve:

1. Sample high proficiency bilinguals + vary immersion
2. Comprehensive bilingual demographics
3. Assess within Linear Mixed Model (direct and interactive effects)

➤ *Language similarity*

Bilingual Demographics

- **Age of Acquisition**

- Categorical?

Continuum towards native-like processing?

- **Proficiency**

- Vocabulary knowledge?
- Measured objectively vs. subjective report

- Opportunities to encode emotion to L2 words
 - Connect to emotional experiences (Dewaele, 2008)

- **Language Use**

- **Language Learning Contexts**

- Family vs. classroom

- Common usage context (ratings of home vs. university words; Imbault et al., 2022)
- Balanced bilinguals (pupil dilation; Thoma, 2023)

- **Immersion**

Bilingual Demographics across Experiments 1-4

	Immersed in L1 (German)		Immersed in L2 (English)	
	Experiment 1	Experiment 2	Experiment 3	Experiment 4
	<i>Valence task</i>	<i>Arousal task</i>	<i>Valence task</i>	<i>Arousal task</i>
<i>n</i>	158	133	50	46
Age	31.25 (11.68)	32.13 (10.65)	43.30 (14.30)	36.07 (10.69)
Education	5.31 (1.39)	5.23 (1.46)	6.44 (1.16)	6.22 (1.17)
Lextale score percentage correct	85.37 (11.68)	85.19 (11.72)	90.93 (8.10)	89.62 (7.94)
L2 Reading proficiency	9.45 (1.17)	9.54 (0.99)	10.06 (0.98)	10.20 (0.81)
L2 Age of Acquisition (first learned)	8.94 (2.65)	9.10 (3.05)	10.56 (4.73)	9.29 (3.37)
L2 use percentage	33.49 (21.26)	34.08 (22.46)	70.36 (19.27)	67.93 (22.67)
L2 family immersion	4.48 (8.48)	2.42 (7.53)	10.33 (11.91)	7.09 (9.40)
L1 Reading proficiency	10.71 (0.57)	10.72 (0.56)	10.54 (1.01)	10.61 (0.86)
L1 Age of Acquisition (first learned)	1.15 (1.40)	1.30 (1.57)	0.92 (1.34)	1.21 (1.40)
L1 use percentage	61.80 (22.67)	59.99 (23.52)	26.11 (16.02)	26.01 (19.28)
L1 family immersion	27.50 (9.56)	28.50 (9.93)	26.22 (10.95)	23.77 (8.30)

Note. Mean (SD) scores for bilingual demographics measured in Experiments 1–4. Education values represent (1) less than high school; (2) high school; (3) professional training; (4) some college; (5) college; (6) some graduate school; (7) Masters; (8) Ph.D./M.D./J.D. Reading proficiency was measured on an 11-point scale. L2 family immersion measured in years.

Problem 2: Measuring Emotion

Complex Tasks = Many co-occurring processes

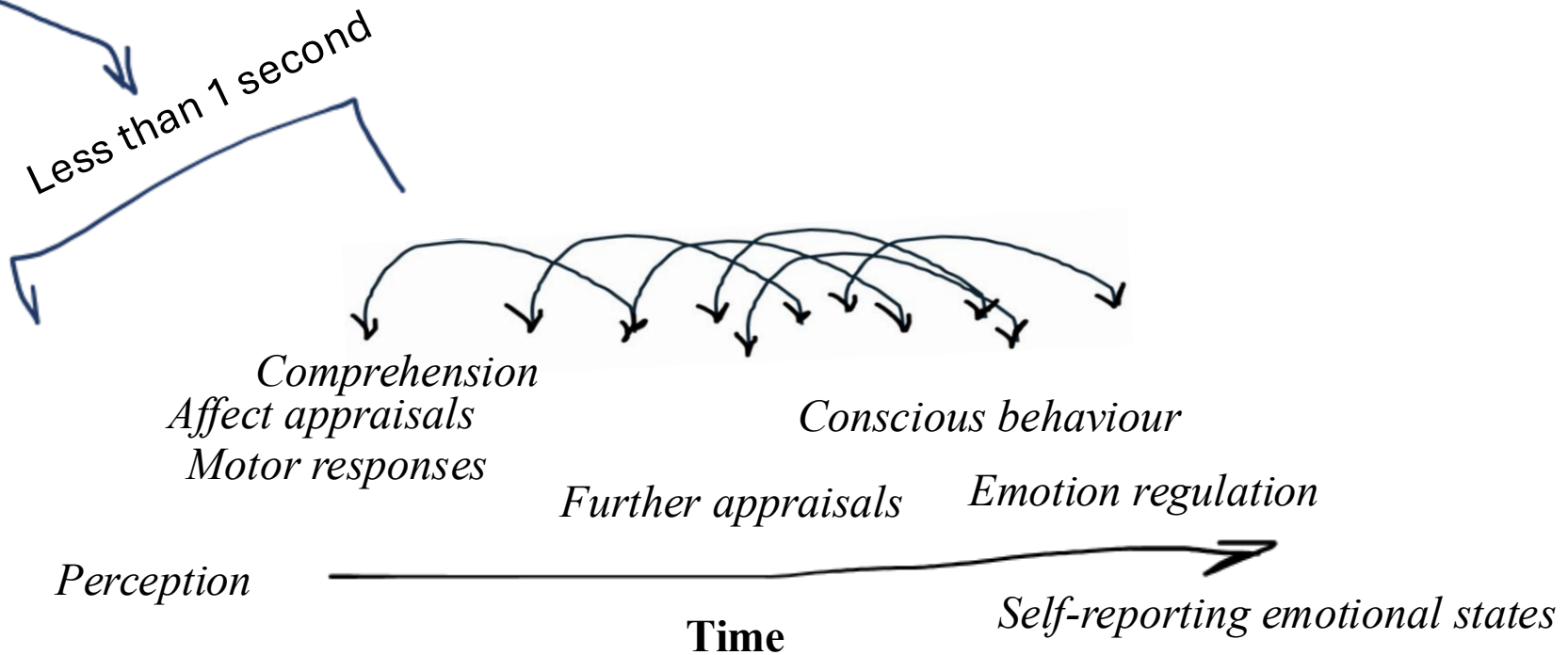
- Simple paradigm: steppingstone to richer context
- Affective appraisals – building blocks
- Isolating emotion (vs. cognitive load, etc.)

Plan to solve:

1. Single words (vs. dilemmas, complex stimuli)
2. Speeded task: Focus on early, automatic affect appraisals
3. Look at interactions (vs. direct effects)

Emotion: Valence & Arousal

- Early affective appraisals (valence, arousal)
- Before more conscious processes occur



Valence x Arousal Interaction

Faster judgements for:
High-arousal negative

Faster judgements for:
Low-arousal positive

Withdraw 

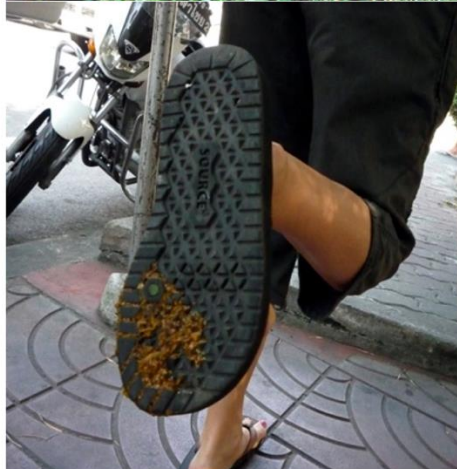
**High
Arousal**



*Incongruent:
Slower RT*

**Negative
Valence**

**Positive
Valence**



*Incongruent:
Slower RT*

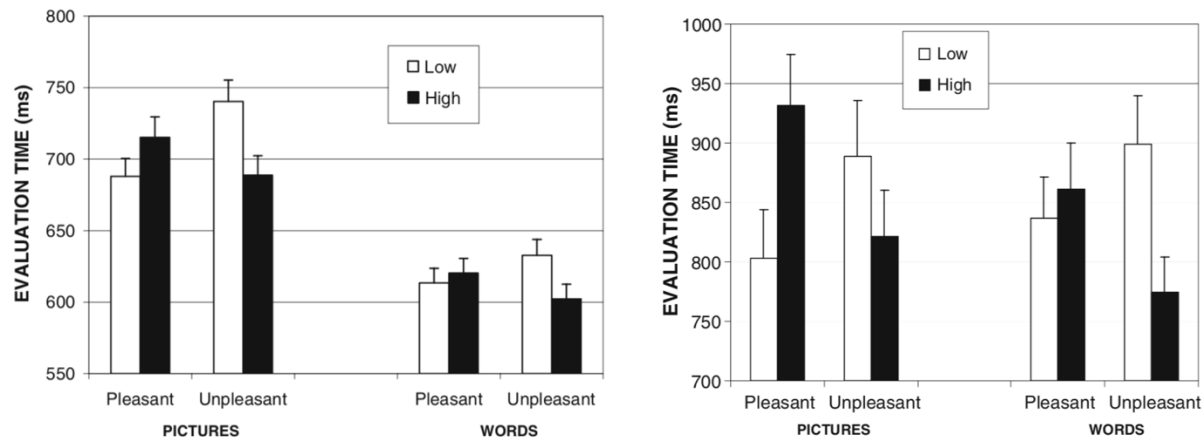
**Low
Arousal**

 **Approach**

Valence x Arousal Interaction

- **Processing pattern** found in **monolingual** research (Purkis et al., 2009; Robinson et al., 2004).
- Considers arousal as an interactive (not additive) factor
- Faster judgements for:
 - **High-arousal negative stimuli** (defensive/withdraw)
 - **Low-arousal positive stimuli** (appetitive/approach)

Not explored in a bilingual context



Mean evaluation time in the evaluation of (picture) stimulus valence and arousal as a function of normative valence and arousal ratings. From "An increase in stimulus arousal has differential effects on the processing speed of pleasant and unpleasant stimuli" by H. Purkis, O. Lipp, M. Edwards, and R. Barnes, 2009, *Motivation & Emotion*, 33, 359

Valence x Arousal Interaction

Valence Arousal Hypothesis (monolingual):

- Robinson et al. (2004): positive vs. negative; pleasant vs. unpleasant feelings; priming an unrelated motor performance tasks
- Wang et al. (2018): related to approach-withdrawal tendencies
- Citron et al. (2013): high-neg and low-pos words elicited larger amplitude ERPs during early lexical processing
- Citron et al. (2014): fMRI shows more cognitive resources required for incongruent words

Valence only: Conflicting Results?

Processing advantages for affective stimuli:

- Processing advantage in the L2 for **positive** over neutral words (e.g. Mergen & Kuruoglu, 2017; Kazanas & Altarriba, 2016)
- Advantage for **positive *and* negative** over neutral words (e.g., Chen et al., 2015; Conrad et al., 2011)
- **No differences** across languages (e.g., Chen et al., 2015; Mergen & Kuruoglu, 2017; Ponari et al., 2015)
- **Inconclusive** results (e.g., Conrad et al., 2011)
- Further interactions (between bilingual ‘type’ and abstractness; Ferré et al., 2018)

Disentangling Emotion Effects

- Interactions with valence and arousal may indicate **emotion processing**
 - Interactions with bilingual moderators may highlight variables more relevant to ‘embodied’ language
- Direct effects on RT may suggest other cognitive processes:
 - **Cognitive load** associated with L2 comprehension
 - **E.G.:** Slower processing overall
 - **E.G.:** Word Length effects

Problem 3: Lexical Variables

- ✓ Word length, word frequency, neighbourhood density
- *Concreteness, word type, emotion-label vs laden*
- ✓ Cognate status

Plan to solve:

1. Control through selection of Stimuli
2. Control within Linear Mixed Model
- *But could not control for all issues*
3. Included pictures: Control condition

Methodology: Stimuli Set

1. Combine Corpora: ANEW and ANGST
(Bradley & Lang, 1999, and Schmidtke et al., 2014)
2. Collect Frequency and Length
SUBTLEX (Brysbaert et al., 2011, 2012)
3. Match pair-wise across languages
 - i. Cross check list-wise
4. Norming preliminary list
 - i. Large number of exclusions based on arousal
5. Cross-linguistic exclusions: cognates, false friends
6. Add orthographic neighbourhood density
CLEARPOND (Marian et al., 2012)

Final Stimuli Set



- 40 words (presented 4x)
- 48 pictures (presented 2x)

• Not perfect ☹️

Issues:

- Concreteness
- Word type
- *Frequency*

Negative words			
<i>Low arousal</i>			<i>High arousal</i>
entfernt	aloof	ärgern	annoy
dunkel	dark	Angriff	assault
unförmig	deformed	Erpressung	blackmail
niedergeschlagen	discouraged	Kakerlake	cockroach
schwach	feeble	zerstören	destroy
teilnahmslos	listless	Furcht	fear
trüb	muddy	Geisel	hostage
starr	rigid	demütigen	humiliate
Harn	urine	Schrei	scream
müde	weary	Spinne	spider

Positive words			
<i>Low arousal</i>			<i>High arousal</i>
Brauch	custom	bewundert	admired ^a
Puppe	doll	erregt	aroused ^a
Fahrstuhl	elevator	Stimmung	cheer
Haarnadel	hairpin	schmücken	decorate
gelenkig	limber	Jubel	joy
ernähren	nourish	lebhaft	lively
Straße	street	Geld	money
sauber	tidy	außerordentlich	outstanding ^a
Baum	tree	Leidenschaft	passion ^a
Fenster	window	strahlend	radiant

Aims & RQs

- Can we replicate the Valence Arousal Conflict Pattern?
- Does the pattern occur across both L1 and L2? **Do bilinguals process emotions similarly?**
- How do bilingual demographics impact processing? **Which type of bilinguals are impacted?**

Study Design

- **Affective judgement tasks**
 - Valence judgement (Exp 1 & 3)
 - Arousal judgment (Exp 2 & 4)
 - Only difference: Labels for sorting
- **Presented in both languages (German-English)**
 - Within-subjects
- **Replicated with different bilingual populations**
 - Immersed in L1 (Exp 1 & 2)
 - Immersed in L2 (Exp 3 & 4)

Procedure

1. L1 task
2. L2 task
3. Picture task
4. LexTale
5. Comprehension check
6. LEAP-Q

Counterbalanced



DV = Reaction Time

Valence
Positive vs. Negative

Arousal
Arousing vs. Calming

L1 dominant: Living in German-speaking countries

L2 dominant: Living in English-speaking countries

Experiment 1

Experiment 2

Experiment 3

Experiment 4

Valence Judgement: Exp 1 & 3

Unpleasant

Pleasant

friend

Practice trials first

Labels switched
sides for half the
presentations

Words presented
in random order

Each word
presented four
times



Valence Judgement: Exp 1 & 3

Negativ

Positiv



Pictures presented
in random order

Each picture
presented two
times



Analysis: Controls and Moderators

Item level:

- Frequency
- Length
- Orthographic neighbourhood density

Participant level:

- Age
- Education

Bilingual Demographics

- Vocabulary knowledge
- Proficiency (Self-reported)
- Age of acquisition
- Family immersion
- Daily Language Exposure

EXP1&3: Valence Judgement

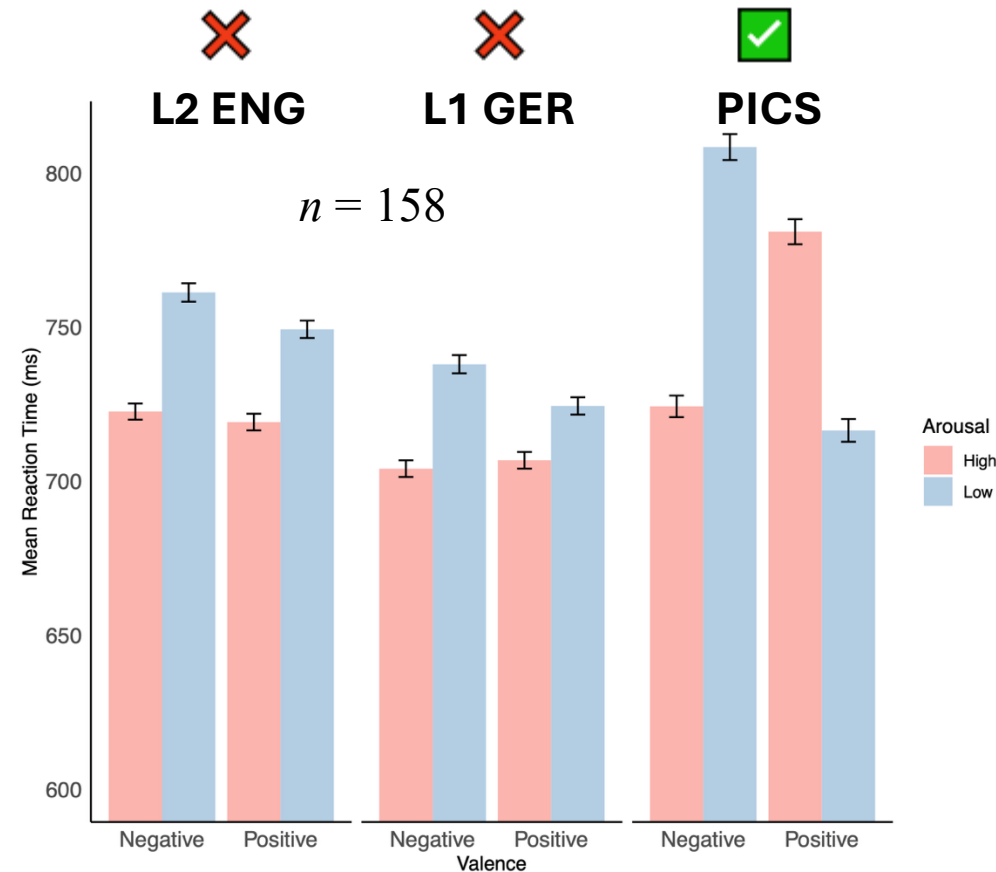
Valence x Arousal interaction
driven by Picture Stimuli

DE EXP1:

- Interaction = pics only
- L1 vs. L2 = same
- Words vs. pictures = diff

GB EXP3:

- Similar results
- Low power ($n = 50$)

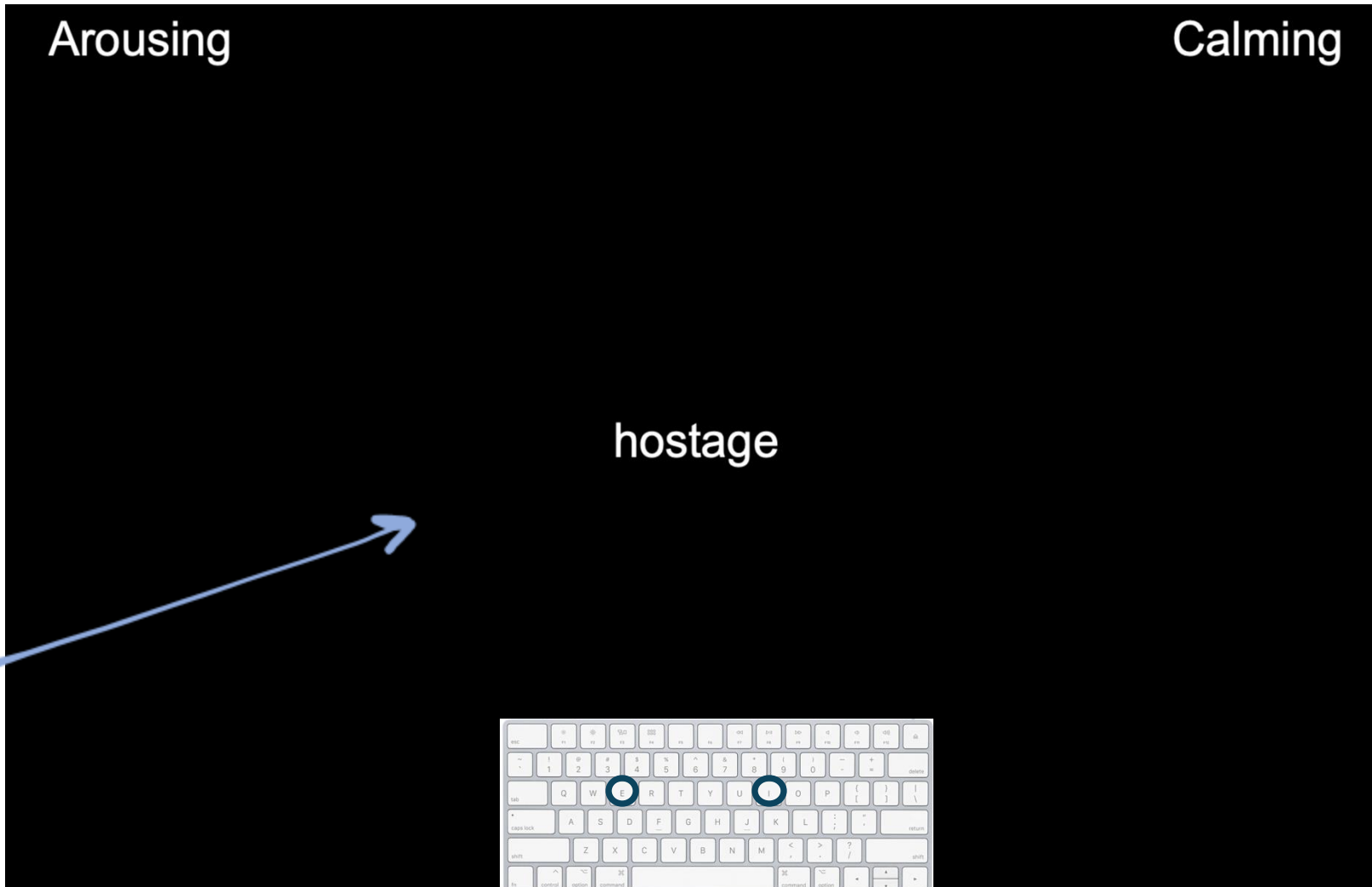


Impact of Frequency

- **Frequency**

- Interacted with **arousal**:
- Low arousal words processed faster when higher frequency
- High arousal words – little difference
- Frequency interaction also found in EXP 4 (arousal judgement)

Arousal Judgement: Exp 2 & 4



Arousal Judgement: Exp 2 & 4

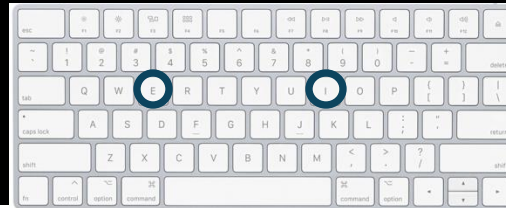
Aufregend

Ruhig



Pictures presented
in random order

Each picture
presented two
times



EXP2&4: Arousal Judgement

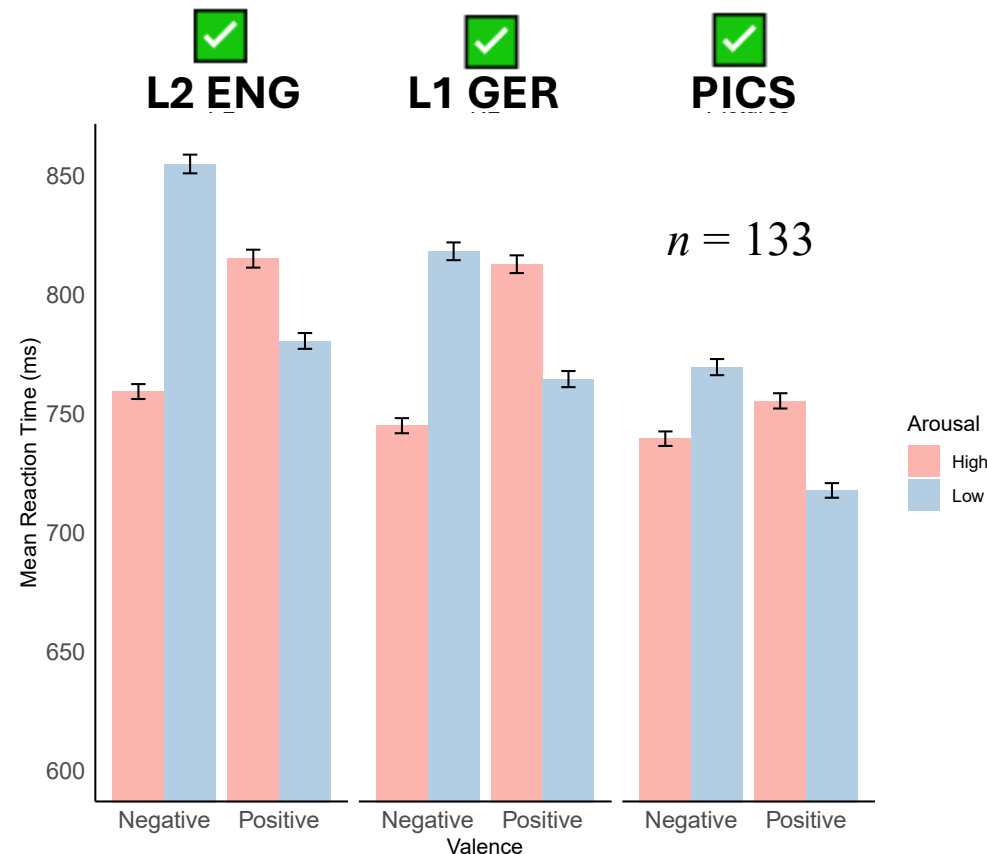
Valence x Arousal interaction
across **ALL** stimuli

DE **EXP2:**

- Interaction = all stimuli
- L1 vs. L2 = same
- Words vs. pictures = diff

GB **EXP4:**

- *Slightly* different – but power too low to speculate ($n = 46$)



Results Overview

Q1: Can we replicate the Valence Arousal Conflict Pattern with word stimuli?

- **Task dependent: Only in the arousal task** 

Task	Immersion Environment	Analysis Outcomes			
		L2 words	L1 words	Picture stimuli	Across all stimuli
Exp 1: Valence judgement	L1 (German)	No	No	Yes	Yes
Exp 2: Arousal judgement	L1 (German)	Yes	Yes	Yes	Yes
Exp 3: Valence judgement	L2 (English)	No	No	No	Yes
Exp 4: Arousal judgement	L2 (English)	Yes	Yes	Yes	Yes

Q2: Does the pattern occur across both L1 and L2?

- **Yes: same pattern occurs across languages**
- **Bilinguals process affect similarly in L1 and L2**

Q3: How do bilingual demographics impact processing?

- **Inconsistently across studies – but all variables had *some* sort of impact**

Results: Bilingual Impact

Moderating influence	Immersed in L1 (German) Environment		Immersed in L2 (English) Environment	
	Exp 1: Valence Judgement	Exp 2: Arousal Judgement	Exp 3: Valence Judgement	Exp 4: Arousal Judgement
Earlier Age of acquisition	 Low arousal words processed faster			
Higher Lextale score				 Negative words processed slower
Higher Subjective proficiency			 Negative and positive words processed faster; steeper slope for negative words	
Higher L2 family immersion		 Negative words processed slower Low arousal words processed slower		
Higher English Exposure		 High arousal words processed slower		 High arousal words processed slower

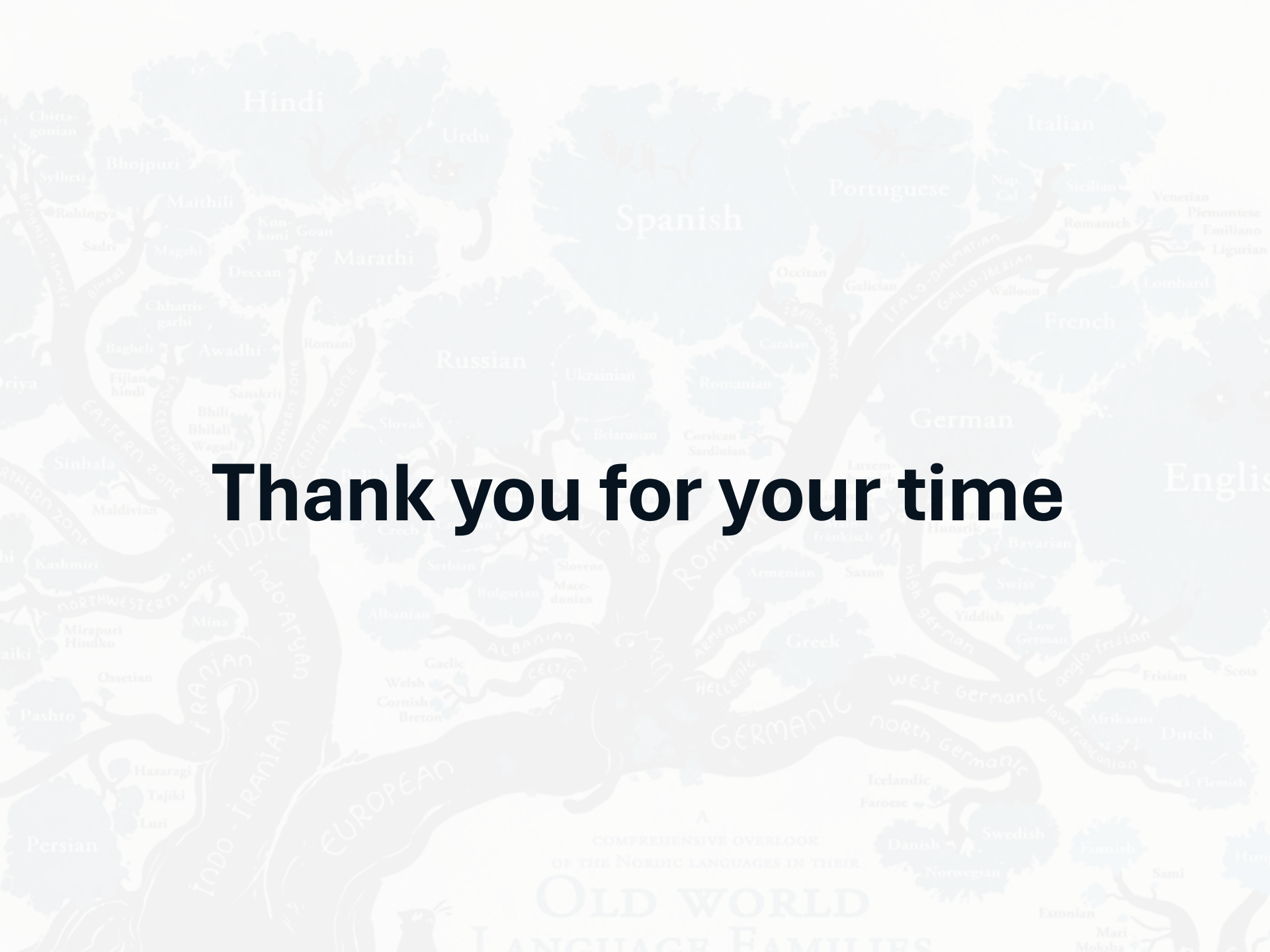
Conclusions

Bilinguals process emotions similarly:

1. For early valence and arousal appraisals
 - Differences may emerge at later levels of processing
2. For highly proficient bilinguals
 - Language experience has an impact
3. For similar language pairs

Arousal Task Effects: Later stage of processing?

- Arousal task – slower and more difficult



Thank you for your time

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